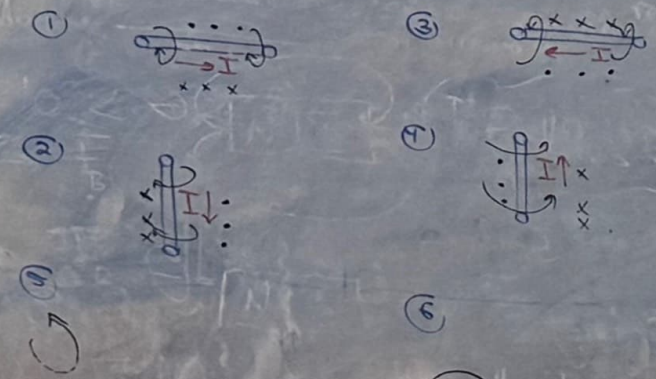


سواء الحقول ايجابية

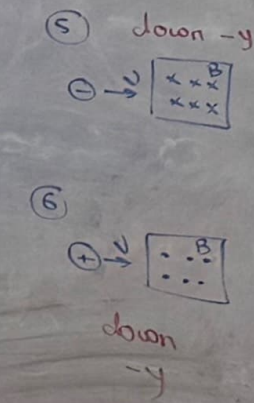
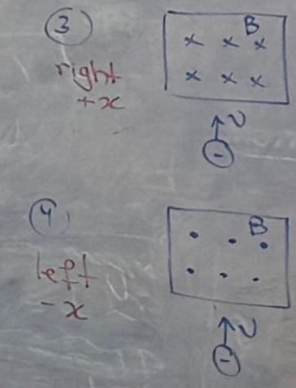
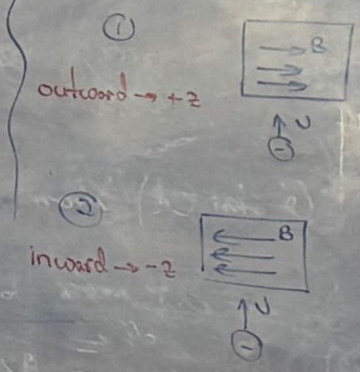
Right Hand Rule: Wire having current (I)



↺ counter clockwise

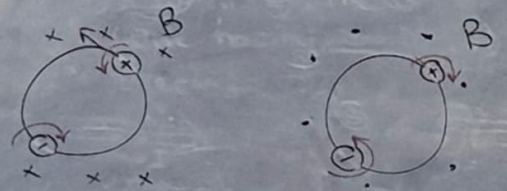
↻ clockwise

Right Hand Rule: moving charge



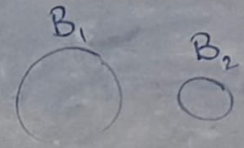
بسم الله الرحمن الرحيم

Circular motion



$$qBr = mv$$

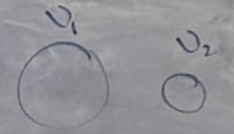
$B \uparrow, r \downarrow$
 $v \uparrow, r \uparrow$



Same speeds

$$B_1 < B_2$$

$$B_2 > B_1$$



Same magnetic fields

$$v_1 > v_2$$

$$v_2 < v_1$$

If B is inward:

- Proton moves counter clockwise
- electron moves clockwise

بسم الله الرحمن الرحيم

$\vec{F}_E$ v_s $\vec{F}_B$

$\vec{F}_E$ is parallel to $\vec{E}$

$\vec{F}_E$ does work

$\vec{F}_E$ acts on (moving-stationary)
charge

- Perpendicular

- does no work

- moving charge